Jiale Shao

Robotics & Mechatronics | Mechanism Design | Embedded Prototyping

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MEng Design Engineering student at Imperial College London who builds working machines end to end: mechanism, circuit, firmware and the physical prototype. Built closed-loop control from scratch on a **self-balancing robot**, led the mechanism design of a robotic planter, and followed design into manufacturing at **Chery Racing Development**; keen to bring hands-on mechatronics and design-for-assembly skills to a robotics team.

EDUCATION

Imperial College London — MEng Design Engineering 2024 – 2028 (Expected)

Modules: Robotics, Electronics & Circuit Design, Computer Science, Solid Mechanics, Finite Element Analysis, Fluid & Thermal Dynamics, Human-Centred Design Engineering

Awards: **WRO (World Robot Olympiad) Provincial Champion** (2023); **School Principal's First-Class Scholarship** (Top 3% in high school, 2021–2024)

ROBOTICS & MECHATRONICS PROJECTS

Self-balancing Segway — DE2 Electronics coursework, pair project 2026

- **Built up** a two-wheeled PyBench robot (MicroPython, MPU6050 IMU, two encoder DC motors) through five open-ended challenges, from IMU tilt-to-drive and beat-synced choreography to a self-balancing inverted pendulum.
- **Implemented** per-wheel **PID speed control** on interrupt-counted encoder pulses (100 ms window) with integral anti-windup and a dead band, holding speed when the wheels were loaded by hand.
- **Wrote** the balance controller as a **200 Hz** loop: a complementary filter (α 0.95) fuses accelerometer pitch with gyro rate, and the D-term uses gyro rate directly to avoid differentiating a noisy angle.
- **Tuned** gains live with a potentiometer, sweeping Kp alone to bound the stable window (−4 too weak, −8 oscillating) before adding I and D (final Kp −10, Ki 40, Kd −1, set point −1.6° to offset the centre of mass).

Hachi-Pot Bionic Smart Planter — Team project, mechanical lead

- **Led** CAD, structure and mechanism for an **ESP32** robot planter that reacts to light, soil moisture and touch with four moods, shown through six mechanical facial expressions.
- **Designed** a cup-in-cup mechanism in Fusion 360: a servo lifts the plant while a stepper indexes the expression wheel behind a static face window; an L298N-driven wheel base lets it move.
- **Recovered** from a low-fidelity prototype with no room for components or wiring: agreed with my teammate to fix the hardware rather than work around it, and redrew all **8 snap-fit, glue-free FDM layers** with tighter tolerances; Rev B assembled cleanly and demoed successfully.

Stackin' Beats — CTO, Imperial team project (4 members) 2026

- **Owned** the circuit, cam mechanism and working prototype of a tactile music kit: 24 ID bricks on a **1-Wire** bus read by an Arduino Mega, and a DC-motor camshaft of 8 snail cams that turns the melody into a moving wave.

PROFESSIONAL EXPERIENCE

Chery Racing Development, Chery Automobile — Aerodynamics Intern Jul – Sep 2026

Global Technology Innovation Center

- **Designed in CAD** the next-generation aero package for a 330 electric performance sedan (Exlantix ES-based), including an **adjustable wing-angle mechanism** and swan-neck rear-wing mounts, against downforce and drag targets I set.
- **Worked with** a CFD colleague on iterative optimisation (CL −0.155 → −0.565) and **followed** first-generation carbon kit manufacturing at the factory, linking design intent to what was built.

ENGINEERING ANALYSIS PROJECTS

Prosthetic Blade Structural Optimisation — Coursework

- **Raised** the first natural frequency from 16 Hz to 51.7 Hz across three geometries in Ansys Workbench; static safety factor 16.6, Gerber fatigue checked. Also: electric GT aero (CD −35% in CFD, wind-tunnel validated).

TECHNICAL SKILLS

Electronics & Control: PID, complementary filtering; C++, Python/MicroPython, MATLAB; Arduino Mega, ESP32, PyBench, 1-Wire, DC / servo / stepper control, TB6612/L298N drivers, sensors, bench testing

CAD / CAE: Fusion 360, SolidWorks, Ansys Mechanical/Workbench, Ansys Fluent

Prototyping & DFMA: FDM/SLA 3D printing, laser cutting, tolerancing, snap-fits, design for assembly. **Languages:** Chinese (native), English (advanced)